



Version 1.03.0

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API Documentation

Setup

UCurses framework setup information can be found in the Quick Start document file.

Namespaces

UCursesInclude
All UCurses classes are under this namespace.

Properties

UCurses Instance
Static property used to access the singleton object from any script in a scene.
Vector2Int GridSize
Property used to get the row and column size of the character grid.
Vector2Int DosScreenResolution
Property used to get the reference resolution the character grid uses.
float AspectRatio
Property used to get the aspect ratio of the displayed image used for the main camera.
Vector2Int CharacterSize
Property used to get the height and width of each character grid spot.
bool OffsetLine
Property used to get if the grid is using a horizontal offset line to increase the width of the character grid after each spot.
FilterMode ScreenFilterMode
Property used to get the texture filtering mode being used on the main camera image.
FilterMode CharacterFilterMode
Property used to get the texture filtering mode being used on each character space on the character grid.

Methods

void setDosScreenMode(DosScreenMode screenMode)		
Description	Sets the DOS console screen settings being emulated within Ucurses.	
Parameters	DosScreenMode screenMode	Takes a DosScreenMode container object with the required information needed to setup the character grid.

void setCharSprites(CharSprite[] charSprites)		
Description	Sets the array of ascii image sprites used for the character grid.	
Parameters	CharSprite[] charSprites	Takes a array of CharSprite container objects used for holding character sprites.

void printCharacter(Vector2Int position, char character, Color32 color)		
Description	Prints a character with a given color on the character grid.	
Parameters	Vector2Int position	Position on the grid where the character will be printed.
	char character	Character to be printed on the grid.
	Color32 color	The color of the printed character.

void printCharactersRectangle(Vector2Int position, char character, Color32 color, int width, int height)		
Description	Prints a rectangle of a character with a given color on the character grid.	
Parameters	Vector2Int position	Position on the grid of the top left most character of the rectangle being printed.
	char character	Character used in the rectangle to be printed on the grid.
	Color32 color	The color of the printed rectangle.
	int width	The width of the printed rectangle.
	int height	The height of the printed rectangle.

void printCharactersRectangle(Vector2Int position, char character, Color32 color, int width, int height, bool filled)		
Description	Prints a rectangle of a character with a given color on the character grid.	
Parameters	Vector2Int position	Position on the grid of the top left most character of the rectangle being printed.
	char character	Character used in the rectangle to be printed on the grid.
	Color32 color	The color of the printed rectangle.
	int width	The width of the printed rectangle.
	int height	The height of the printed rectangle.
	bool filled	If true the rectangle is filled, if false it is an outline.

void printString(Vector2Int position, string charString, Color32 color)		
Description	Prints a string of a characters with a given color on the character grid.	
Parameters	Vector2Int position	Position on the grid of the left most character of the string being printed.
	string charString	String to be printed on the grid.
	Color32 color	The color of the printed string.

void printNumberInt(Vector2Int position, int numericValue, Color32 color)		
Description	Prints a whole number with a given color on the character grid.	
Parameters	Vector2Int position	Position on the grid of the left most character of the number being printed.
	int numericValue	Int value to be printed to the grid.
	Color32 color	The color of the printed number.

void printNumberInt(Vector2Int position, int numericValue, Color32 color, uint paddingLength, bool addZeros)		
Description	Prints a whole number with a given color on the character grid.	
Parameters	Vector2Int position	Position on the grid of the top left most space of the rectangle being printed.
	int numericValue	Int value to be printed to the grid.
	Color32 color	The color of the printed number.
	uint paddingLength	How many padding spaces in front of the number are used.
	bool addZeros	If true adds zero characters in unused padding spaces.

void printNumberDouble(Vector2Int position, double numericValue, Color32 color)		
Description	Prints a decimal number with a given color on the character grid.	
Parameters	Vector2Int position	Position on the grid of the left most character of the decimal number being printed.
	double numericValue	Double value to be printed to the grid.
	Color32 color	The color of the printed decimal number.

void printNumberDouble(Vector2Int position, double numericValue, Color32 color, uint paddingLength, uint decimalLength, bool addZeros)		
Description	Prints a decimal number with a given color on the character grid.	
Parameters	Vector2Int position	Position on the grid of the left most character of the number being printed.
	double numericValue	Double value to be printed to the grid.
	Color32 color	The color of the printed number.
	uint paddingLength	How many padding spaces in front of the decimal place are used.
	uint decimalLength	How many padding spaces in back of the decimal place are used.
	bool addZeros	If true adds zero characters in unused padding spaces in front.

void printBackground(Vector2Int position, Color32 color)		
Description	Changes the background color of a character grid space.	
Parameters	Vector2Int position	Position on the grid where the background will be printed.
	Color32 color	The color of the printed background.

void printBackgroundRectangle(Vector2Int position, Color32 color, uint width, uint height)		
Description	Changes the background color of a rectangle of character grid spaces.	
Parameters	Vector2Int position	Position on the grid where the background will be printed.
	Color32 color	The color of the printed background.
	int width	The width of the rectangle being printed.
	int height	The height of the rectangle being printed.

void printBackgroundRectangle(Vector2Int position, Color32 color, int width, int height, bool filled)		
Description	Changes the background color of a rectangle of character grid spaces.	
Parameters	Vector2Int position	Position on the grid of the top left most space of the rectangle being printed.
	Color32 color	The color of the printed background.
	int width	The width of the rectangle being printed.
	int height	The height of the rectangle being printed.
	bool filled	If true the rectangle is filled, if false it is an outline.

Vector2Int mousePosition()	
Description	Retrieves the position of the mouse on the character grid.
Return	Returns what character grid position the mouse is currently over. If the mouse is outside the grid will return (-1, -1).

void characterBlink(Vector2Int position, bool enable)		
Description	Sets if character on the character grid is blinking.	
Parameters	Vector2Int position	Position on the grid where the character is/will be blinking. By default the blink is set to 0.5 seconds.
	bool enable	If true blinking is turned on, if false blinking is turned off.

void characterBlink(Vector2Int position, bool enable, float seconds)		
Description	Sets if character on the character grid is blinking.	
Parameters	Vector2Int position	Position on the grid where the character is/will be blinking.
	bool enable	If true blinking is turned on, if false blinking is turned off.
	float seconds	Sets the blink rate for the character in seconds.

void characterUnderlined(Vector2Int position, bool enable)		
Description	Sets if character on the character grid is underlined.	
Parameters	Vector2Int position	Position on the grid where the character is/will be underlined.
	bool enable	If true underline is turned on, if false underline is turned off.

void characterUnderlined(Vector2Int position, bool enable, uint width)		
Description	Sets if character on the character grid is blinking.	
Parameters	Vector2Int position	Position on the grid where the character is/will be underlined.
	bool enable	If true underline is turned on, if false underline is turned off.
	uint width	How grid spaces from the starting position the underline will go to the right.

string unixString(string stringValue)		
Description	Converts a string to use Unix style (LF) newline characters. All string based methods use this style.	
Parameters	string stringValue	The string to be converted.
Return	string	The converted string.

void printCharArray(Vector2Int position, char[,] charArray, Color32 color)		
Description	Prints a 2D array of characters with a given color on the character grid.	
Parameters	Vector2Int position	Position on the grid where the character is/will be underlined.
	char[,] charArray	The char array that you want to print to the string.
	Color32 color	The color of the printed char array.

void printCharArray(Vector2Int position, char[,] charArray, Color32[,] colorArray)		
Description	Prints a 2D array of characters with a corresponding 2D Color32 array on the character grid.	
Parameters	Vector2Int position	Position on the grid where the character is/will be underlined.
	char[,] charArray	The char array that you want to print to the string.
	Color32[,] colorArray	An array that holds the color for each char in the char array.

Code Page 437 Character Set

Name	Char	ASCII Name	Char	ASCII
SOH (Start of Header)	☺	1 Slash	/	47
STX (Start of Text)	☹	2 Zero	0	48
ETX (End of Text)	♥	3 One	1	49
EOT (End of Transmission)	♦	4 Two	2	50
ENQ (Enquiry)	♣	5 Three	3	51
ACK (Acknowledge)	♠	6 Four	4	52
BEL (Bell)	•	7 Five	5	53
BS (BackSpace)	▣	8 Six	6	54
HT (Horizontal Tabulation)	○	9 Seven	7	55
LF (Line Feed)	▣	10 Eight	8	56
VT (Vertical Tabulation)	♂	11 Nine	9	57
FF (Form Feed)	♀	12 Colon	:	58
CR (Carriage Return)	♪	13 Semicolon	;	59
SO (Shift Out)	♪	14 Less than	<	60
SI (Shift In)	☀	15 Equals sign	=	61
DLE (Data Link Escape)	▶	16 Greater than	>	62
DC1 (Device Control 1)	◀	17 Question mark	?	63
DC2 (Device Control 2)	↕	18 At	@	64
DC3 (Device Control 3)	!!	19 Upper case A	A	65
DC4 (Device Control 4)	¶	20 Upper case B	B	66
NAK (Negative Acknowledge)	§	21 Upper case C	C	67
SYN (Synchronous Idle)	—	22 Upper case D	D	68
ETB (End of Transmission Block)	↕	23 Upper case E	E	69
CAN (Cancel)	↑	24 Upper case F	F	70
EM (End of Medium)	↓	25 Upper case G	G	71
SUB (Substitute)	→	26 Upper case H	H	72
ESC (Escape)	←	27 Upper case I	I	73
FS (File Separator)	└	28 Upper case J	J	74
GS (Group Separator)	↔	29 Upper case K	K	75
RS (Record Separator)	▲	30 Upper case L	L	76
US (Unit Separator)	▼	31 Upper case M	M	77
Space		32 Upper case N	N	78
Exclamation mark	!	33 Upper case O	O	79
Quotation Mark	"	34 Upper case P	P	80
Hash	#	35 Upper case Q	Q	81
Dollar	\$	36 Upper case R	R	82
Percent	%	37 Upper case S	S	83
Ampersand	&	38 Upper case T	T	84
Apostrophe	'	39 Upper case U	U	85
Open bracket	(	40 Upper case V	V	86
Close bracket	)	41 Upper case W	W	87
Asterisk	*	42 Upper case X	X	88
Plus	+	43 Upper case Y	Y	89
Comma	,	44 Upper case Z	Z	90
Dash	-	45 Open square bracket	[	91
Full stop	.	46 Backslash	\	92

Name	Char	ASCII Name	Char	ASCII	
Close square bracket	]	93	Lower case i with grave	ì	141
Caret	^	94	Upper case A with diaeresis	Ä	142
Underscore	_	95	Upper case A with ring above	Å	143
Grave accent	`	96	Upper case E with acute	É	144
Lower case a	a	97	Lower case ae	æ	145
Lower case b	b	98	Upper case AE	Æ	146
Lower case c	c	99	Lower case o with circumflex	ô	147
Lower case d	d	100	Lower case o with diaeresis	ö	148
Lower case e	e	101	Lower case o with grave	ò	149
Lower case f	f	102	Lower case u with circumflex	û	150
Lower case g	g	103	Lower case u with grave	ù	151
Lower case h	h	104	Lower case y with diaeresis	ÿ	152
Lower case i	i	105	Upper case O with diaeresis	Ö	153
Lower case j	j	106	Upper case U with diaeresis	Ü	154
Lower case k	k	107	Cent sign	¢	155
Lower case l	l	108	Pound sign	£	156
Lower case m	m	109	Yen sign	¥	157
Lower case n	n	110	Peseta sign	₧	158
Lower case o	o	111	Lower case f with hook	ƒ	159
Lower case p	p	112	Lower case a with acute	á	160
Lower case q	q	113	Lower case i with acute	í	161
Lower case r	r	114	Lower case o with acute	ó	162
Lower case s	s	115	Lower case u with acute	ú	163
Lower case t	t	116	Lower case n with tilde	ñ	164
Lower case u	u	117	Upper case N with tilde	Ñ	165
Lower case v	v	118	Feminine ordinal indicator	ª	166
Lower case w	w	119	Masculine ordinal indicator	º	167
Lower case x	x	120	Inverted question mark	¿	168
Lower case y	y	121	Reversed not sign	¬	169
Lower case z	z	122	Not sign	¬	170
Open brace	{	123	Vulgar fraction one half	½	171
Pipe		124	Vulgar fraction one quarter	¼	172
Close brace	}	125	Inverted exclamation mark	¡	173
Tilde	~	126	Left-pointing double angle quotation mark	«	174
Delete	△	127	Right-pointing double angle quotation mark	»	175
Upper case C with cedilla	Ç	128	Light shade	░	176
Lower case u with diaeresis	ü	129	Medium shade	▒	177
Lower case e with acute	é	130	Dark shade	▓	178
Lower case a with circumflex	â	131	Box drawings light vertical		179
Lower case a with diaeresis	ä	132	Box drawings light vertical and left	├	180
Lower case a with grave	à	133	Box drawings vertical single and left double	┤	181
Lower case a with ring above	å	134	Box drawings vertical double and left single	├	182
Lower case c with cedilla	ç	135	Box drawings down double and left single	┐	183
Lower case e with circumflex	ê	136	Box drawings down single and left double	┐	184
Lower case e with diaeresis	ë	137	Box drawings double vertical and left	├	185
Lower case e with grave	è	138	Box drawings double vertical		186
Lower case i with diaeresis	ï	139	Box drawings double down and left	┐	187
Lower case i with circumflex	î	140	Box drawings double up and left	┐	188

Name	Char	ASCII Name	Char	ASCII	
Box drawings up double and left single	⌞	189	Greek lower case phi	φ	237
Box drawings up single and left double	⌟	190	Greek lower case epsilon	ε	238
Box drawings light down and left	⌋	191	Intersection	∩	239
Box drawings light up and right	⌌	192	Identical to	≡	240
Box drawings light up and horizontal	⌍	193	Plus-minus sign	±	241
Box drawings light down and horizontal	⌎	194	Greater-than or equal to	≥	242
Box drawings light vertical and right	⌏	195	Less-than or equal to	≤	243
Box drawings light horizontal	⌐	196	Top half integral	∫	244
Box drawings light vertical and horizontal	⌑	197	Bottom half integral	∫	245
Box drawings vertical single and right double	⌒	198	Division sign	÷	246
Box drawings vertical double and right single	⌓	199	Almost equal to	≈	247
Box drawings double up and right	⌔	200	Degree sign	°	248
Box drawings double down and right	⌕	201	Bullet operator	·	249
Box drawings double up and horizontal	⌖	202	Middle dot	·	250
Box drawings double down and horizontal	⌗	203	Square root	√	251
Box drawings double vertical and right	⌘	204	Superscript lower case n	n	252
Box drawings double horizontal	⌙	205	Superscript two	²	253
Box drawings double vertical and horizontal	⌚	206	Black square	■	254
Box drawings up single and horizontal double	⌛	207			
Box drawings up double and horizontal single	⌜	208			
Box drawings down single and horizontal double	⌝	209			
Box drawings down double and horizontal single	⌞	210			
Box drawings up double and right single	⌟	211			
Box drawings up single and right double	⌠	212			
Box drawings down single and right double	⌡	213			
Box drawings down double and right single	⌢	214			
Box drawings vertical double and horizontal single	⌣	215			
Box drawings vertical single and horizontal double	⌤	216			
Box drawings light up and left	⌥	217			
Box drawings light down and right	⌦	218			
Full block	▀	219			
Lower half block	▁	220			
Left half block	▂	221			
Right half block	▃	222			
Upper half block	▄	223			
Greek lower case alpha	α	224			
Lower case sharp s	ß	225			
Greek upper case letter gamma	Γ	226			
Greek lower case pi	π	227			
Greek upper case letter sigma	Σ	228			
Greek lower case sigma	σ	229			
Micro sign	μ	230			
Greek lower case tau	τ	231			
Greek upper case letter phi	Φ	232			
Greek upper case letter theta	Θ	233			
Greek upper case letter omega	Ω	234			
Greek lower case delta	δ	235			
Infinity	∞	236			

Change Log

1.0

Initial release.

1.01

Character sprites changed from individual files to a sprite map to improve performance.

1.02

Added multi-line support functionality for `printString()`.

Added `unixString()` and `printCharArray()` methods.

1.02.1

Fixed bug with `mousePosition()` giving out of bounds coordinates if the mouse was on the edge of the screen.

1.03.0

Fixed bug with `mousePosition()` giving out of bounds coordinates if the mouse was on the edge of the screen.

1.03.0

Added horizontal alignment for screen output and offset in setup.

Changed `mousePosition()` to use screen coordinates instead of world coordinates.

Fixed unnecessary position calculation on tile creation.

Fixed deprecation warning regarding `FindFirstObjectByType`.